GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY
FACULTY OF COMPUTING
DEPARTMENT OF INFORMATION TECHNOLOGY
INTAKE 41



ESSENTIALS OF ARTIFICIAL INTELLIGENCE
PROJECT PROPOSAL

	Group Details		
Group Number	**Degree Programme**	**Student Number**	**Student Name**
38	DBA	D/DBA/25/0024	KMGPD Kariyawasam
	BIT	D/BIT/24/0019	KHTU Jayasekara
	BSE	D/BSE/25/0014	PV Kavinda
	BIT	D/BIT/24/0013	WGISK Wedage
Project Title	AI-Based Herbal Plant Identification and Digital Herbarium System - HerbSense		

Abstract

Medicinal plants remain central to traditional healthcare, yet identifying them from appearance alone is difficult, and look-alike species can be harmful when mistaken for one another. This report presents **HerbSense**, a web application that identifies medicinal plants from a photograph and communicates how far the answer can be trusted. It pairs a YOLO11n deep-learning model, deployed as a Roboflow serverless workflow, with a knowledge-based layer. The model returns its top three candidates with confidence scores. When the best score is 75% or higher the result is shown directly; otherwise a rule-based expert system asks up to three plant-feature questions (leaf arrangement, leaf shape, crushed-leaf scent, growth habit) and fuses the answers with the image score using 0.6 × image score + 0.4 × feature-match score. The final result adds a digital herbarium record, look-alike comparisons and a severity-graded safety warning, with an optional language-model description.

The system uses React, a Node.js/Express proxy that keeps API keys off the browser, and PostgreSQL. All 15 automated tests pass. A simulation over 132 ordered plant pairs showed that a wrong first guess was corrected in 100% of cases with all four correct answers and 80.3% with the scent answer alone. Limitations include the 12-plant herbarium, uncalibrated thresholds and the lack of an independent evaluation of the image model.

1. Introduction

The World Health Organization reports that traditional medicine remains an important source of primary healthcare in many countries (WHO, 2013). In South Asia, plants such as neem, tulsi, turmeric and gotu kola are part of everyday Ayurvedic practice, and using them safely depends on knowing which plant is in hand.

Computer vision can identify a plant from a phone photo, but a bare label such as "Tulsi – 62%" leaves open what a person should do when the model is unsure. **HerbSense** follows a *guided identification pipeline*: it shows several candidates, asks for extra evidence when confidence is low, checks the answer against curated botanical knowledge and ends with a safety message proportionate to the remaining uncertainty. It consists of a React front end, a Node.js/Express back end that proxies the YOLO11n workflow and serves a 12-plant digital herbarium, an optional language-model description service and a PostgreSQL database. It is an educational tool, not a diagnostic or prescribing system.

2. Problem Background

Medicinal plants are associated with traditional healthcare knowledge, but their appearance can be difficult for a beginner to interpret. Different plants may share a leaf shape, while the same species may look different depending on age, growing conditions or the part photographed. A photograph can also hide useful details such as leaf arrangement, growth form or scent.

Computer vision offers a way to recognise visual patterns from images. However, a model may return a plausible prediction even when available evidence is weak. An identification interface therefore needs to communicate uncertainty and provide a way to inspect alternatives. HerbSense combines photo analysis with stored botanical features and user observations to address this need.

The project also organises plant information in a digital herbarium. This connects the predicted name with scientific names, plant families, habitat, commonly recorded uses, active compounds and precautions. Such information supports learning; it does not establish that a plant is safe or that a claimed use is clinically effective.

3. Problem Statement

Users need a practical way to identify medicinal plants from photographs while understanding uncertainty and possible confusion with similar species. A system that presents only one label may provide insufficient support when the image is unclear or candidate plants have similar appearances. The project problem is to develop a web application that combines image predictions with additional plant-feature evidence, retrieves relevant botanical information and communicates limitations through clear warnings.

4. Objectives

The aim is to design, implement and test a guided, safety-aware identification system. The specific objectives are:

1. Detect and classify medicinal plants from a photo with a deep-learning detector.
2. Show the top three plants and classify confidence as high, medium or low.
3. Verify or adjust uncertain predictions with rule-based questions and transparent score fusion.
4. Provide herbarium information, look-alike comparisons and a severity-graded safety warning.
5. Add an optional LLM description that cannot influence the identification.
6. Build a secure full-stack application with accounts and scan history.

5. Selected AI Techniques

HerbSense is a hybrid system: a learned perception model, symbolic knowledge-based components and a generative model used only for explanation (Table 2).

Table 2: AI techniques used in HerbSense

Technique	Category	Role	Technique
Computer vision and YOLO11n deep learning	Deep learning	Detects and classifies the plant; returns class and confidence.	YOLO11n detector
Confidence policy	Decision rules	High ≥ 0.75, medium 0.45–0.74, low < 0.45; decides whether to ask questions.	Confidence policy
Rule-based verification and fusion.	Knowledge-based AI	Selects discriminating feature questions and scores the answers.	Rule-based expert system
Weighted evidence fusion	Hybrid AI	s = 0.6 × image score + 0.4 × rule score.	Weighted evidence fusion
Similarity retrieval	Knowledge-based	Ranks look-alikes by curated links (10), family (3) and shared features (1 each).	Similarity retrieval
Language model	Generative AI	Writes a short plant description; never used for the decision.	Language model

Computer vision and YOLO11n deep learning
YOLO11n is the image-recognition model used through Roboflow. YOLO is a family of neural-network models for computer vision, and the nano variant is a relatively small model. It's learned visual features support recognition of plant classes from images. The backend sends an image to the hosted workflow and receives predicted class labels and confidence scores.

At the application level, this operation is inference: an existing deployed model processes a new image. The backend extracts predictions from the workflow response, removes duplicate class entries by keeping the highest score, and sorts candidates. The project does not train a model during a user scan.

Confidence policy
Threshold rules classify scores as high, medium or low. A high initial score allows the workflow to proceed directly to a result. Lower scores lead to verification when useful questions are available. Warning severity depends on the final score and available reference information.

Rule-based verification and fusion.
The verification layer uses predefined botanical knowledge. Each herbarium record stores accepted feature values, and the rule engine compares user answers with those values. The four available feature categories are leaf arrangement, leaf shape, crushed-leaf scent and growth habit. Invalid answers and "Not sure" responses are excluded from scoring.

Weighted evidence fusion
The system combines image confidence with a rule score, calculated as matched answers divided by comparable answered questions:
Final score = 0.6 × image score + 0.4 × rule score

Candidates are reranked using this score. Without comparable answers, the image score is retained. The weights are design choices, and the final score is not a validated probability of correctness.

Language model.
The plant name, never the image, is sent to an OpenAI-compatible endpoint (default gpt-4o-mini) for a 90 - 120 word description shown as supplementary notes.

6. Justification of AI Techniques

YOLO11n supports the central requirement of recognizing plants from photographs. Using a hosted Roboflow workflow allows the application to request inference without running the image model on the user's device. YOLO11 supports computer-vision tasks and has a compact nano variant, although project-specific speed and accuracy still require measurement.

The rule-based system adds information that may be unclear or absent in a photograph. Its stored feature values can be inspected and corrected, and the reasoning can be explained through answer matches. This suits a small herbarium because the verification knowledge can be maintained directly without training another model.

The language model improves readability by producing a short description. Keeping it separate from identification prevents generated text from directly changing the predicted species. The reference herbarium and generated descriptions should remain distinguishable because their information sources differ.

7. System Design

HerbSense uses a three-tier architecture (Figure 1). The browser never contacts Roboflow or the LLM provider; the Express back end holds the credentials, rate-limits requests (60 per minute per IP) and applies the verification logic. Identification still works if the description model or database fails, and every result carries a warning.

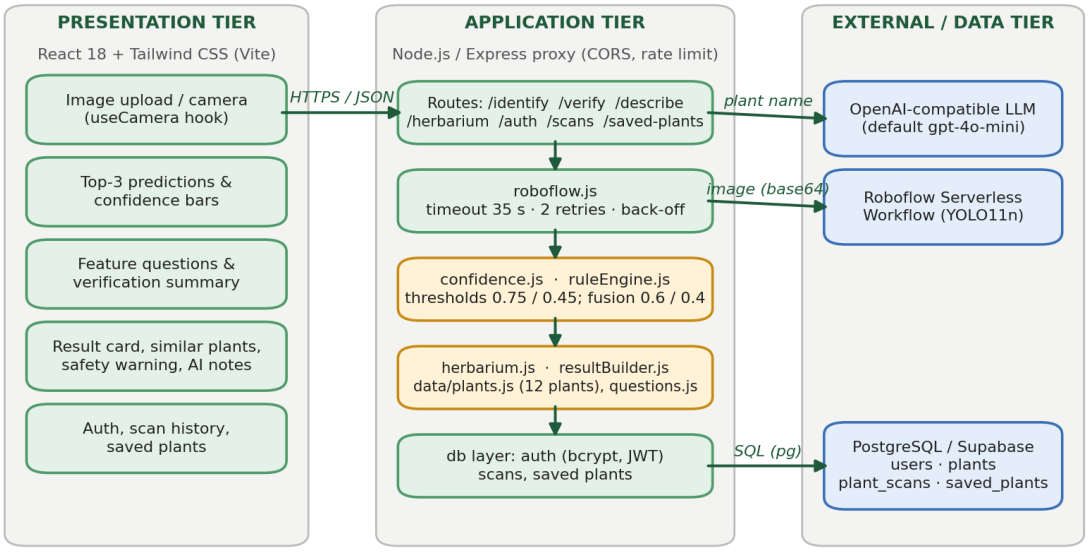


Figure 1: HerbSense system architecture

Figure 2 shows the guided pipeline. `POST /api/identify` returns stage `none`, `verification` or `final`; if questions are needed, `POST /api/verify` takes the candidates and answers and returns the final result. A high-confidence result carries an info warning, a medium one "caution – confirm before use", and a low one "critical – do not use this plant".

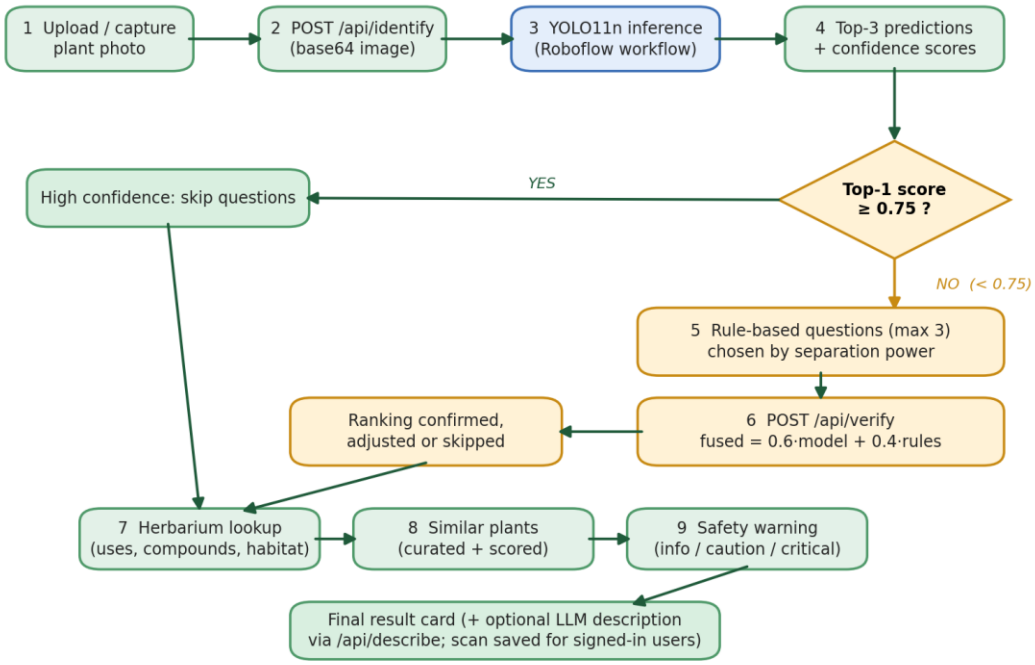


Figure 2: Guided identification pipeline

Table 3: Confidence levels and system behaviour

Level	Score	Behaviour	Warning severity
High	75% or more	Final result immediately.	Info
Medium	45% – 74%	Questions, then result.	Caution – confirm before use
Low	Below 45%	Questions, then result.	Critical – do not use

The digital herbarium holds 12 plants (aloe vera, neem, tulsi, turmeric, ginger, ashwagandha, brahmi, moringa, gotu kola, peppermint, lemongrass and curry leaf) with names, aliases such as "kohomba", family, uses, compounds, habitat, precautions, feature values and look-alike tips. No two plants share all four feature values. PostgreSQL stores `users` (bcrypt hashes), `plants`, `plant_scans` and `saved_plants`. Security measures include server-side API keys, restricted CORS, bcrypt with 12 salt rounds, 30-day JWTs and validation of all inputs.

8. Implementation Details

8.1 Frontend

The interface is built with React, Vite and Tailwind CSS. Components handle image upload, camera capture, top predictions, verification questions, plant information, similar plants and warning messages. React hooks track states such as analysing, verifying, submitting, final, no result and error. Account components support registration, login, profiles and scan-history views

8.2 Backend and model integration

Node.js and Express expose REST endpoints. POST /api/identify sends the image for inference, while POST /api/verify processes feature answers. GET /api/herbarium and its name-based route return reference information. POST /api/describe generates a botanical description. Other routes provide authentication, profiles, database plants, scans and saved plants.

The Roboflow client uses a server-side API key and a 35-second request timeout. It retries selected network and server failures, and maps common response codes to clearer error messages. Response parsing accommodates different workflow output formats. Scores are clamped to the range from zero to one before ranking.

8.3 Verification and result construction

The rule engine selects questions, validates answers, computes match scores and reranks candidates. The result builder attaches herbarium information, similar plants and warning messages. Unknown plant records produce a result without reference details and with a warning explaining the missing herbarium information.

8.4 Descriptions and persistence

Descriptions are cached in memory for 24 hours. Requests for the same plant can share an in-progress request, which reduces duplicate calls. The description client applies timeouts and retries for temporary service failures. PostgreSQL is accessed through a connection pool and parameterised SQL queries. Identification can be initiated in guest mode, while account features support personal activity records.

8.5 Security controls

Passwords are hashed using bcrypt, and authentication uses JSON Web Tokens. The server validates selected request fields, applies CORS origin rules and limits API requests to 60 per minute per IP address. JSON requests have a 20 MB body limit. API credentials belong on the server and should be supplied through environment configuration.

These controls do not by themselves establish production security. Deployment needs an explicitly configured JWT secret, verified database TLS settings and consistent authentication checks for scan creation and history access. Submitted user identifiers should not substitute for authenticated ownership.

9. Testing and Results

Automated tests. The offline suite (`npm run test:unit`, with a stubbed Roboflow client) has 15 tests covering confidence thresholds, herbarium lookup, question selection, answer sanitising, ranking adjustment, the result builder and five API route scenarios. All 15 passed in about 0.37 seconds.

Simulation of the verification layer. For each ordered pair of herbarium plants (T, D) (132 pairs), the wrong plant D was ranked first (0.55) and the true plant T second (0.40). A simulated user answered with T's accepted values and the real `verifyCandidates` function re-ranked the list (Table 3, Figure 3). In a control with the correct plant first, it stayed first in 132 of 132 cases. Scent alone is weakest because five plants share "none" and three share "spicy". These results show the logic works under idealised conditions (correct answers, true plant in the top three); they do not measure real-world accuracy.

Table 3: Correction of a wrong first guess (132 cases per row)

Answers supplied	Corrected	Still wrong	Success rate
All four features	132	0	100.0%
Leaf arrangement + scent	128	4	97.0%
Leaf shape + growth habit	120	12	90.9%
Scent only	106	26	80.3%

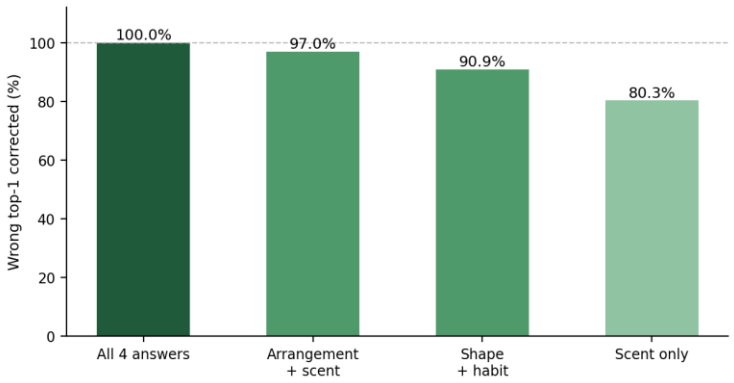
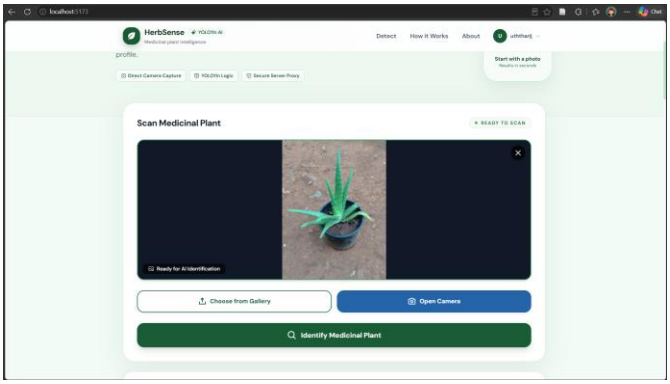
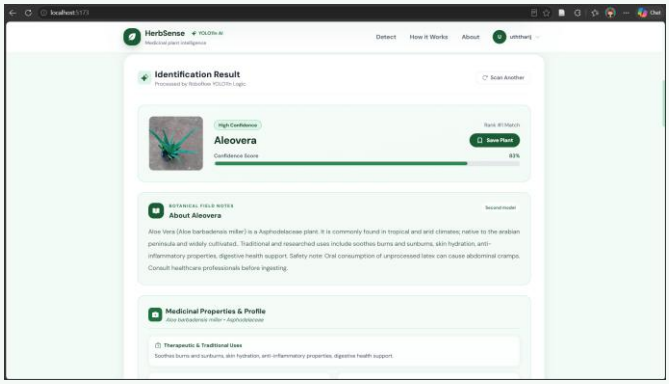
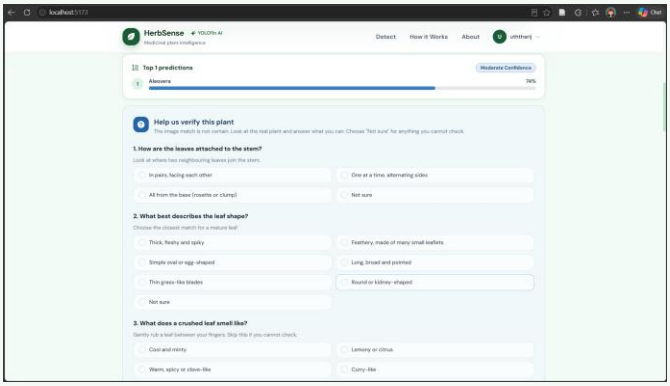
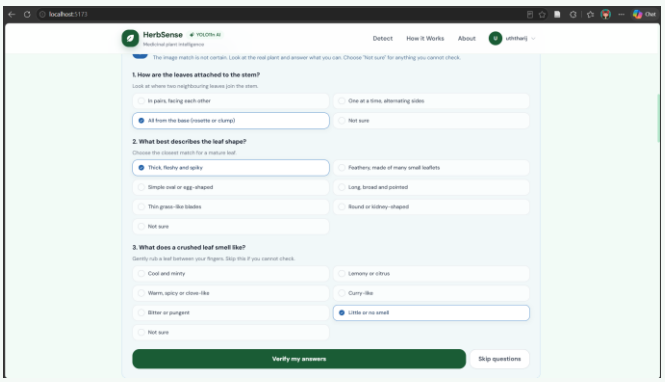
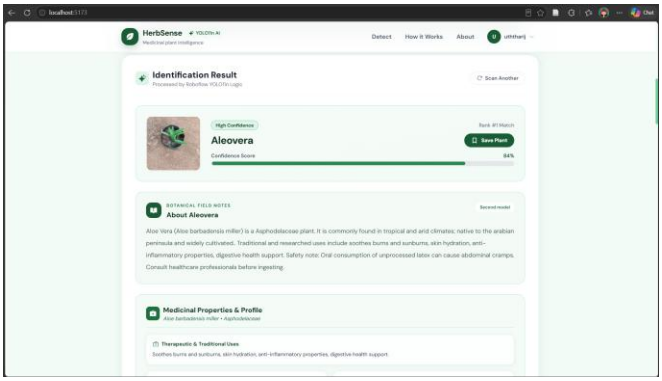
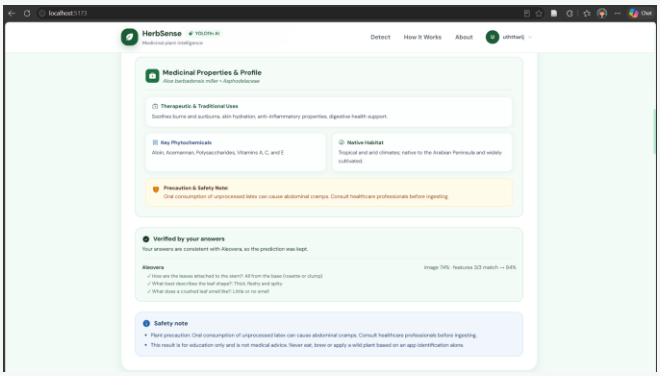
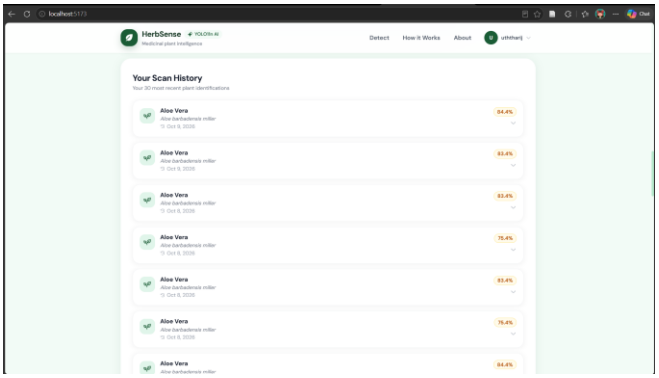


Figure 3: Share of wrong first predictions corrected, by answers supplied

9.1 Screenshots to Insert

Insert the following six screenshots after the relevant paragraphs in Sections 8 and 9. Use actual system screens and values. Crop unnecessary browser space and hide account details or credentials. Replace each instruction with its image and caption before submission.

Screenshot and required evidence	Screenshot
Image upload or camera screen with the selected plant photo.	
Uncertain prediction with candidate scores and the feature-question form.	 

Screenshot and required evidence	Screenshot
	
Result after verification, showing confirmed or adjusted ranking and feature-match explanation.	
Final result with herbarium details, similar plants and safety warning.	
Signed-in scan history showing a stored identification record.	

10. Limitations and Discussion

The verification herbarium contains 12 plants and four main feature categories. It does not cover the full variety of medicinal plants or dangerous look-alikes. Identification quality can change with image lighting, focus, background, viewpoint and plant condition. Independent image-model accuracy has not yet been established.

Verification depends on correct user observations and accurate stored features. A user may misunderstand a question or select the wrong value. The rule engine only reranks candidates returned to it; it cannot recover the correct plant if that species is absent from the candidate list. Skipped questions provide no additional evidence.

The confidence thresholds and fusion weights are manually defined and have not been calibrated on held-out data. A higher combined score therefore does not prove that a prediction is reliable. Feature matching also gives equal importance to comparable questions even when some features may be more dependable than others.

Training settings, dataset split information and model evaluation outputs need to accompany the deployed model to make the image-recognition component reproducible. The herbarium count must not be treated as the model training-class count. Similarly, the number of stored profiles does not describe the number of training images.

Generated descriptions may contain incorrect or unsupported information. The description request is not grounded in retrieved herbarium references. Internet and provider availability affect inference and description services. Security hardening, multilingual access and systematic user testing also remain important before public deployment.

Future work should use independent field photos to evaluate image predictions and tune thresholds and weights. The herbarium can be expanded with botanist-reviewed sources, additional look-alikes and coordinated database updates. Sinhala and Tamil interfaces, clearer question guidance and source-grounded descriptions would improve accessibility and information quality.

11. Ethical and Social Considerations

An incorrect plant identification can lead to harmful use. HerbSense therefore presents warnings and should be used for education rather than diagnosis, treatment or deciding whether a plant can be consumed. High confidence is not proof of safety. Medical and botanical information should be reviewed by qualified professionals before users act on it.

Traditional uses should be distinguished from evidence of clinical effectiveness. Herbarium entries need traceable references, and traditional knowledge should be represented respectfully with appropriate attribution. Generated text should be identifiable as AI-produced and checked before it is relied on.

Images and account details may contain personal information. Users should understand that images are sent to an external inference service and that activity records may be stored. Data collection should be limited to what the service needs, with appropriate access controls, retention rules and deletion options.

Testing should include different environments, devices and users. Limited class coverage or image diversity can make performance uneven. Accessible wording and future Sinhala and Tamil support can help more users understand the questions and warnings. Educational use should also avoid encouraging unnecessary harvesting of rare plants or disturbance of natural habitats.

12. Conclusion

HerbSense implements a guided medicinal plant identification workflow using YOLO11n, rule-based verification and weighted evidence fusion. The application presents candidate predictions, asks distinguishing feature questions when needed, retrieves digital herbarium information and compares similar plants. A separate language model generates botanical descriptions, while account and database features support scan records and saved plants.

All 15 automated pipeline tests passed. The reported verification simulation shows that correct feature answers can improve candidate rankings in controlled cases. These results support the verification layer, while real-image accuracy, confidence calibration and user reliability remain to be evaluated. The project provides a working foundation for an educational identification system that communicates uncertainty and makes its verification steps visible.